

Telecom Customer Churn Analysis

1. Project Overview

This project analyzes **telecom customer churn** using customer service, demographic, contract, and billing data from **7,043 customers**. The goal is to identify **churn patterns, customer characteristics, service preferences, and factors associated with customer attrition** to support better customer retention strategies.

2. Dataset Summary

- **Rows:** 7,043
- **Columns:** 21
- **Key Features:**
 - **Customer demographics:** Gender, Senior Citizen, Partner, Dependents
 - **Customer tenure & services:** Tenure, Phone Service, Multiple Lines, Internet Service
 - **Additional services:** Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, Streaming Movies
 - **Contract & billing:** Contract, Paperless Billing, Payment Method, Monthly Charges, Total Charges
 - **Target variable:** Churn
- **Missing Data:** No missing values in the dataset
- **Analysis Tools:** Python, Pandas, NumPy, Matplotlib, Seaborn

3. Exploratory Data Analysis using Python

We performed data cleaning and exploratory analysis using **Pandas, NumPy, Matplotlib, and Seaborn**.

- **Data Loading:** Loaded the customer churn dataset using Pandas

- **Initial Inspection:** Used `df.head()` and `df.info()` to understand the dataset structure, columns, data types, and non-null values.

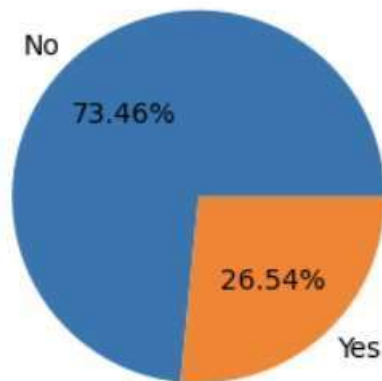
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                7043 non-null   object
2   SeniorCitizen         7043 non-null   int64
3   Partner               7043 non-null   object
4   Dependents            7043 non-null   object
5   tenure                7043 non-null   int64
6   PhoneService          7043 non-null   object
7   MultipleLines         7043 non-null   object
8   InternetService       7043 non-null   object
9   OnlineSecurity        7043 non-null   object
10  OnlineBackup          7043 non-null   object
11  DeviceProtection      7043 non-null   object
12  TechSupport           7043 non-null   object
13  StreamingTV           7043 non-null   object
14  StreamingMovies       7043 non-null   object
15  Contract              7043 non-null   object
16  PaperlessBilling       7043 non-null   object
17  PaymentMethod         7043 non-null   object
18  MonthlyCharges        7043 non-null   float64
19  TotalCharges          7043 non-null   object
20  Churn                 7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

- **Data Cleaning:** Replaced blank values in `TotalCharges` with 0 and converted the column to numeric format.
- **Missing Value Check:** Checked for missing values using `isnull()`.
- **Statistical Analysis:** Used `df.describe()` to understand the numerical data.

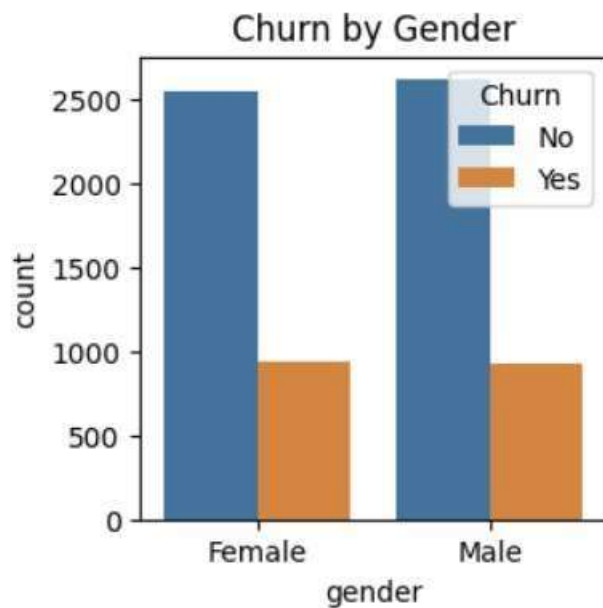
	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

- **Duplicate Check:** Checked customerID for duplicate customer records.
- **Data Transformation:** Converted SeniorCitizen values from **0/1 to No/Yes** for easier understanding.
- **Overall Churn Analysis:** Visualized the number and percentage of churned and retained customers. **26.54% of customers have churned.**

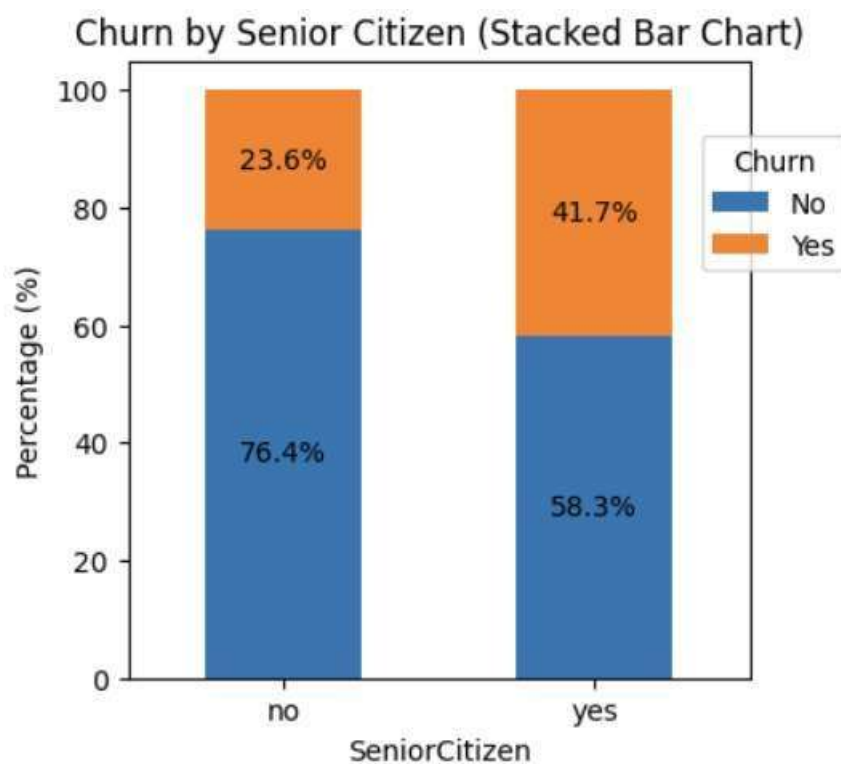
Percentage of Churned Customeres



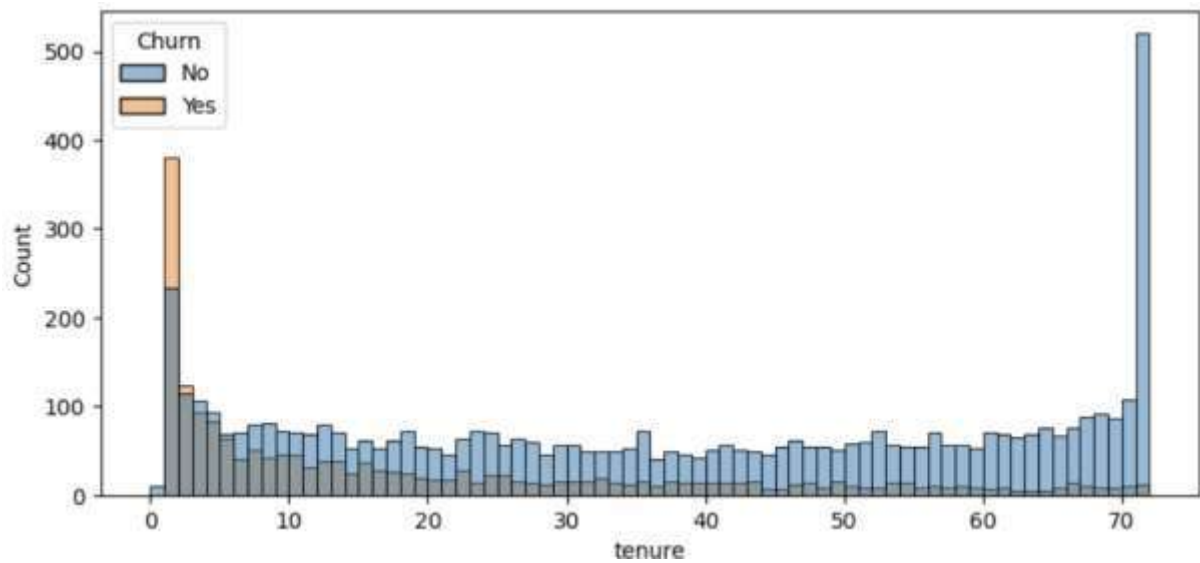
- **Gender Analysis:** Compared churn between male and female customers.



- **Senior Citizen Analysis:** Compared churn patterns between senior citizens and non-senior citizens.



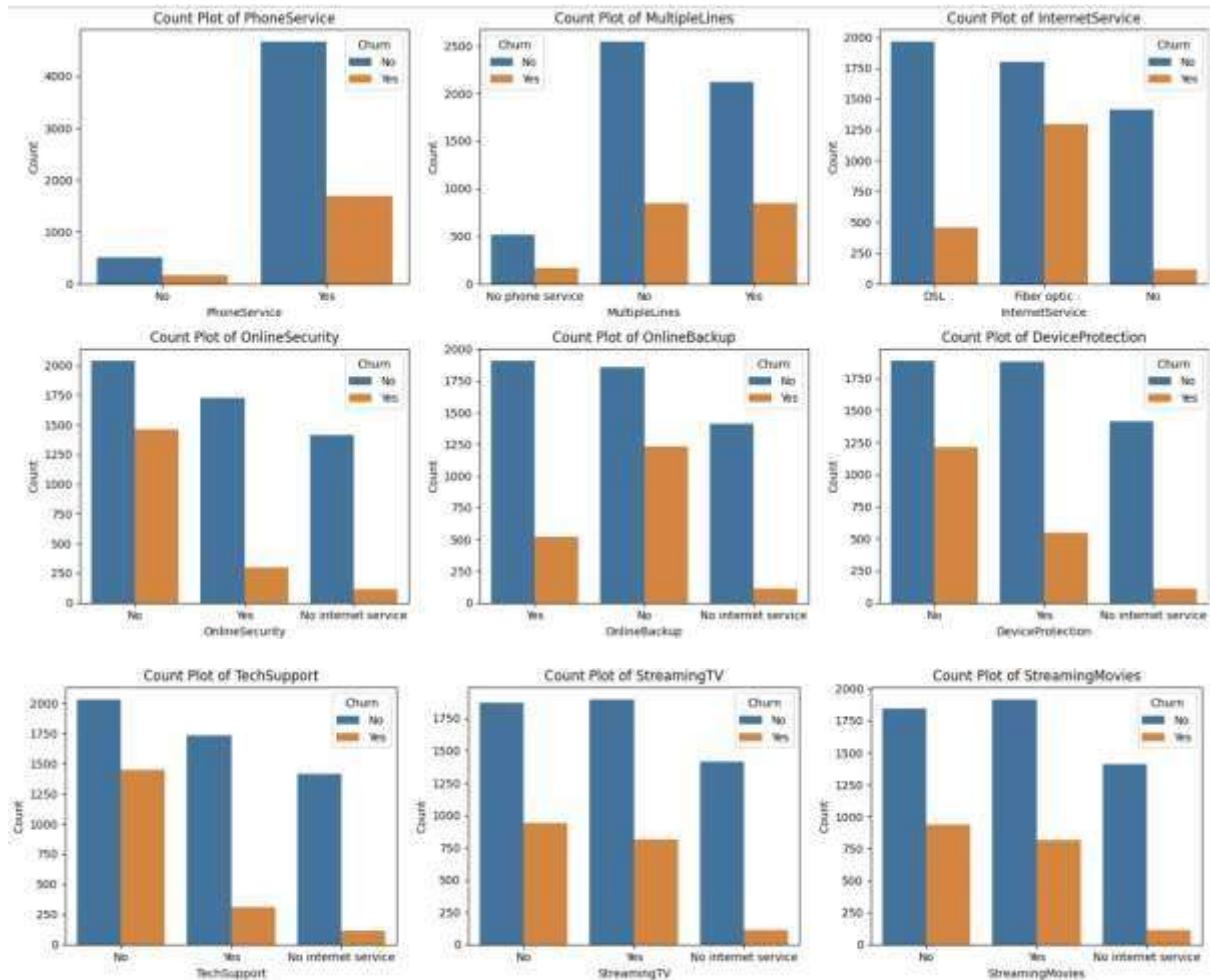
- **Tenure Analysis:** Analyzed how customer tenure is related to churn. Customers with longer tenure generally show better retention.



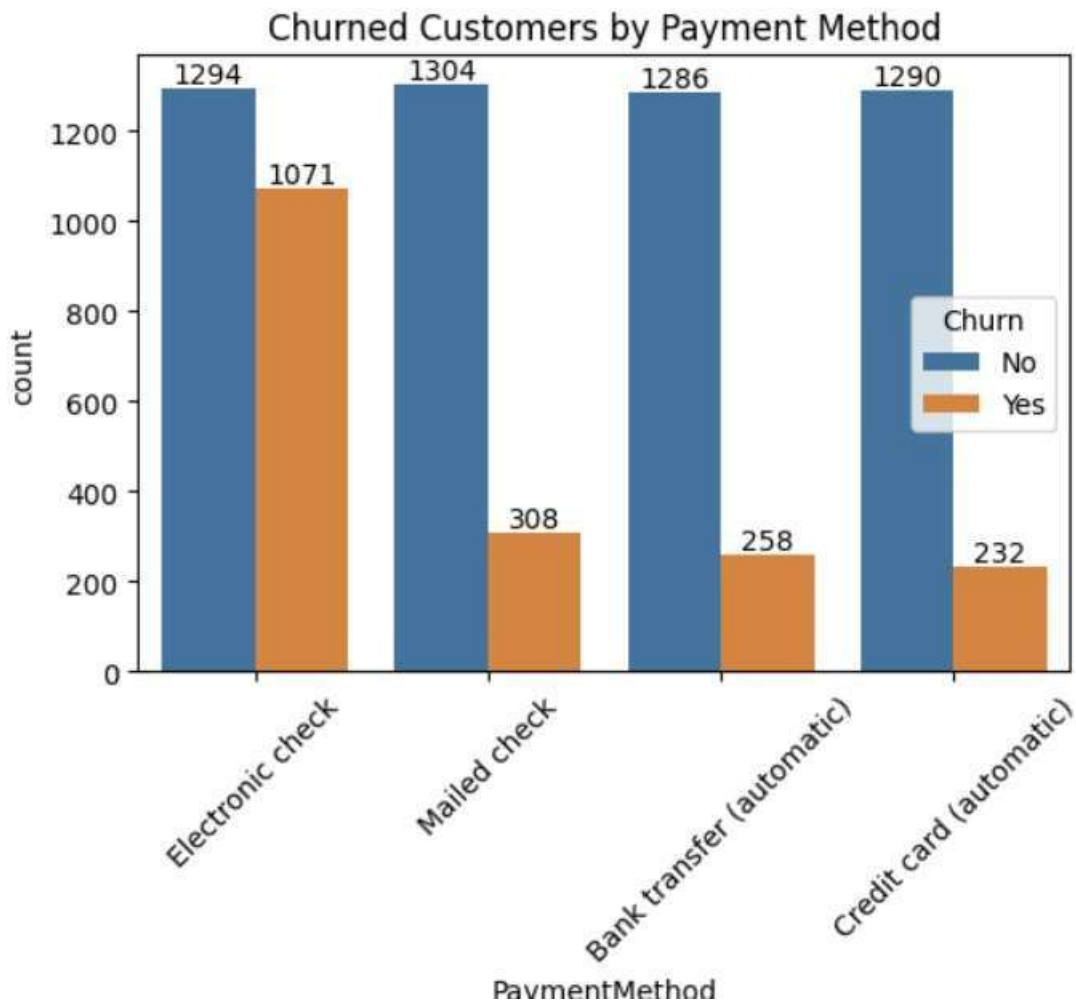
- **Contract Analysis:** Found that **month-to-month customers have higher churn** compared with one- and two-year contract customers.



- **Service Analysis:** Compared churn across different telecom services to identify service-related churn patterns.



- **Payment Method Analysis:** Found that customers using **electronic check** show higher churn compared with other payment methods.



Key Insights & Findings:

- **Contract Type and Churn:**

- Customers on **month-to-month** contracts exhibit the highest churn rate, with **42%** of such customers likely to churn.

- In contrast, customers on **one-year** and **two-year** contracts have churn rates of **11%** and **3%**, respectively.

- **Implication:** Longer contract periods serve as a strong retention tool, as customers with extended commitments are far less likely to leave.

- **Payment Methods and Churn:**

- Customers paying via **electronic checks** show the highest churn rate at **45%**, while those using **credit cards, bank transfers, or mailed checks** have significantly lower churn rates, averaging around **15-18%**.

- **Implication:** The convenience, security, and trust issues related to electronic payments might be contributing factors. Encouraging customers to switch to more stable payment methods could reduce churn.

- **Churn by Tenure:**

- Customers with **less than one year** of tenure are the most likely to churn, with a **50%** churn rate. Those with **1-3 years** of tenure show a decreasing churn trend at **35%**, while customers who have been with the company for **more than three years** have a churn rate of just **15%**.

- **Implication:** Engaging customers early in their journey, especially within the first year, is critical for retention.

- **Churn by Internet Service Type:**

- Customers using **Fiber Optic services** show a higher churn rate of **30%**, compared to **DSL customers** with a churn rate of **20%**.

- **Implication:** This could be due to increased competition or dissatisfaction with service quality. Understanding customer satisfaction with service speed and reliability may help retain fiber optic users.

- **Senior Citizens and Churn:**

- The analysis reveals that senior citizens (aged 65+) have a churn rate of 41%, compared to a 26% churn rate among non-senior citizens.

- **Implication:** Special retention programs and targeted customer service for senior customers may help reduce churn in this demographic.

Recommendations:

- **Promote Long-Term Contracts:** Offer incentives for customers to commit to longer contracts to reduce churn.

- **Address Payment Method Concerns:** Implement campaigns encouraging customers to switch from electronic checks to more reliable payment methods.

- **Customer Engagement in Early Tenure:** Focus on improving the customer experience within the first year, as churn is highest in this period.

- **Special Senior Citizen Retention Programs:** Create personalized offers or assistance programs to retain the senior demographic.